

- 2 (a) Express $\frac{2x^3 + 7x^2 + 5x + 6}{x(x^2 + 2)}$ as the sum of a polynomial and a proper fraction. [1]

$$\begin{array}{r} 2 \\ x^3 + 2x \overline{) 2x^3 + 7x^2 + 5x + 6} \\ \underline{-(2x^3 + 4x)} \\ 7x^2 + x + 6 \end{array}$$

$$\therefore \frac{2x^3 + 7x^2 + 5x + 6}{x(x^2 + 2)} = 2 + \frac{7x^2 + x + 6}{x(x^2 + 2)} \quad (AI)$$

- (b) Hence, express $\frac{2x^3 + 7x^2 + 5x + 6}{x(x^2 + 2)}$ in partial fractions. [5]

$$\text{Let } \frac{7x^2 + x + 6}{x(x^2 + 2)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 2} \quad (MI)$$

$$7x^2 + x + 6 = A(x^2 + 2) + x(Bx + C)$$

When $x = 0$,

$$A(0 + 2) = 6$$

$$2A = 6$$

$$A = 3 \quad (MI)$$

By comparing coeff of

$$x^2: A + B = 7$$

$$3 + B = 7$$

$$B = 4 \quad (MI)$$

$$x: C = 1 \quad (MI)$$

$$\therefore \frac{2x^3 + 7x^2 + 5x + 6}{x(x^2 + 2)} = 2 + \frac{3}{x} + \frac{4x + 1}{x^2 + 2} \quad (AI)$$